Energy Management Scanned Supplement

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

Important: This document is fictional, independently created for academic evaluation, and contains no confidential OEM data or reproduced AUTOSAR standard text.

Field	Value
Document ID	SYN-HLD-EM-001
Version	1.0
Status	Synthetic scanned test artifact
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Scanned component supplement

This document intentionally contains image-only pages to validate automatic OCR routing. It defines the synthetic EnergyManagementCoordinator component.

2. Component definition

Field	Value
Component	EnergyManagementCoordinator
Component ID	SWC-EM-001
Provided interface	If_EnergyMode
Required interfaces	If_BatteryState, If_EngineState
Runnable	REvaluateEnergyMode
Trigger	100 ms timing event
Output states	NORMAL, CONSERVATION, CRITICAL

3. Data elements

Data element	Type	Meaning
EnergyMode	Enumeration	Current energy-management operating mode
LoadSheddingLevel	Unsigned integer	Requested load-shedding level from 0 to 3
EnergyModeValidity	Boolean	TRUE when required inputs are fresh

4. Behaviour

EnergyManagementCoordinator shall publish CONSERVATION when BatteryStateOfCharge_pct is below 25%. It shall publish CRITICAL when BatteryStateOfCharge_pct is below 15% or when If_BatteryState is stale for more than 500 ms.

5. OCR-specific evaluation statement

Expected OCR phrase: EnergyManagementCoordinator publishes If_EnergyMode every 100 ms. This exact sentence is included to support deterministic OCR accuracy testing.

6. Known missing definition

If_BatteryState is intentionally referenced but not defined in the remaining synthetic documents. The system should flag it as a missing interface definition.